

ALVEOLATES

Alveolates are united by possession of the same anatomical quirk – a fringe of tiny sacs around the cell called alveoli, after which the clade is named.

Three groups of superficially different but still single-celled protists are included in the alveolates: the dinoflagellates, the ciliates, and the apicomplexans. Predatory dinoflagellates live in the ocean, most using two whip-like flagella that emerge from grooves in their cell coat at right angles to one another to swim. Some can discharge stinging barbs to immobilize their prey. Others release poisons, and sudden blooms of dinoflagellates are responsible for toxic red tides in parts of the world. A few are bioluminescent, sparkling with light when disturbed at night.

Most ciliates feed on smaller organisms. Their smooth graceful motion is due to the coordinated beating of countless tiny hairs called cilia, which can cover their entire single-celled body. They also use these cilia to waft their food into a furrow that approximates to a mouth. Ciliates are virtually ubiquitous; some even live in the stomachs of mammalian herbivores, where they help in digesting the tough cellulose in plants.

In contrast, all apicomplexans are parasites. They are named for an arrangement of structures, called the apical complex, which helps them penetrate the living cells of animals, from which they derive their nourishment. Infamous among the apicomplexans are the malaria parasites, which penetrate the red blood cells of animals for food, destroying the cells in the process.

DOMAIN	EUKARYOTA
KINGDOM	PROTOCTISTA
CLADE	ALVEOLATA
FAMILIES	222
SPECIES	About 20,000

DEBATE PROTOZOANS

Early classifications placed food-eating microbes into different classes of the animal phylum Protozoa. Some microbes, such as the ciliates and the dinoflagellates, are now grouped together in modern protist taxonomy, although the exact nature of their relationships is debatable.

STALKED CILIATE

Vorticella sp.

This organism has an inverted bell-shaped body on a stalk, which is known to coil like a spring when stimulated.

50–160 μ m



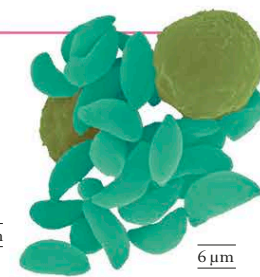
35–90 μ m



40–110 μ m



50–130 μ m



6 μ m

TRUMPET CILIATE

Stentor muelleri

This algae-feeding species, with a horn-shaped cell body, is large for a single-celled organism.

2–3 mm / $\frac{1}{16}$ in

SOIL CILIATE

Colpoda inflata

This normally kidney-shaped species plays an important role in soil ecology but is vulnerable to pesticides.

POND CILIATE

Colpoda cucullus

Usually found in fresh water among decaying plants, this ciliate has food-containing organelles called vacuoles within its cell.

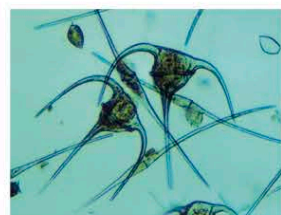
GUT CILIATE

Balantidium coli

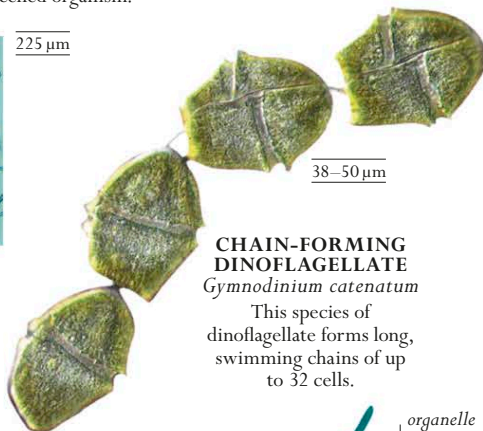
This is the only ciliate known to parasitize humans. Infection can cause intestinal ulcers, or serious intestinal infections.

TOXOPLASMA GONDII

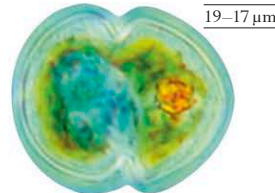
Passing between cats and other mammals (including humans), this apicomplexan causes the disease toxoplasmosis, which is dangerous to the unborn child in pregnancy.



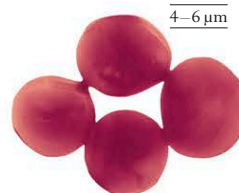
225 μ m



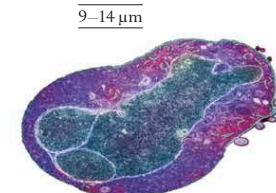
38–50 μ m



19–17 μ m



4–6 μ m



9–14 μ m

THREE-HORNED DINOFLAGELLATE

Tripos muelleri

Formerly known as *Ceratium tripos* and among the most distinctive dinoflagellates, blooms of this plankton can cause dangerous red tides.

CHAIN-FORMING DINOFLAGELLATE

Gymnodinium catenatum

This species of dinoflagellate forms long, swimming chains of up to 32 cells.

MAHOGANY-TIDE DINOFLAGELLATE

Karlodinium veneficum

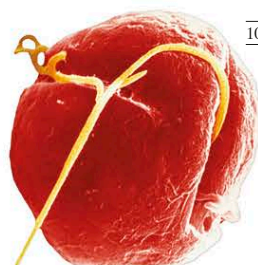
This dinoflagellate plankton can undergo a population explosion leading to "mahogany tides" that are lethal to fish.

CRYPTOSPORIDIUM PARVUM

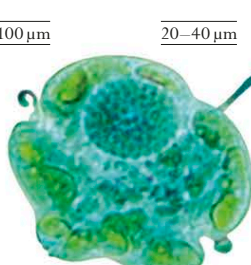
This apicomplexan can cause cryptosporidiosis, a diarrhoeal disease transmitted by ingestion of its spores via faeces, usually in contaminated water.

PLASMODIUM FALCIPARUM

The most deadly of the malaria-causing *Plasmodium* species, this apicomplexan parasite is responsible for the deaths of over 1 million people a year worldwide.



10–100 μ m



20–40 μ m



200–2,000 μ m



40–74 μ m

AKASHIWO SANGUINEA

A large pentagonal species responsible for several recorded instances of harmful algal blooms, this dinoflagellate can both photosynthesize and prey on other plankton.

SEASPARKLE

Noctiluca scintillans

This bioluminescent, plankton-forming species has a gas bag allowing it to float just below the surface.



11–24 μ m

AMPHIDIINIUM CARTERAE

This plankton-forming species causes ciguatera, a fish poisoning that can cause human fatalities if contaminated fish are consumed.